LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 16: Customer Demographics Analysis

Aim

To visualize customer demographics using a bar chart, pie chart, table, and an interactive Tableau dashboard.

Dataset

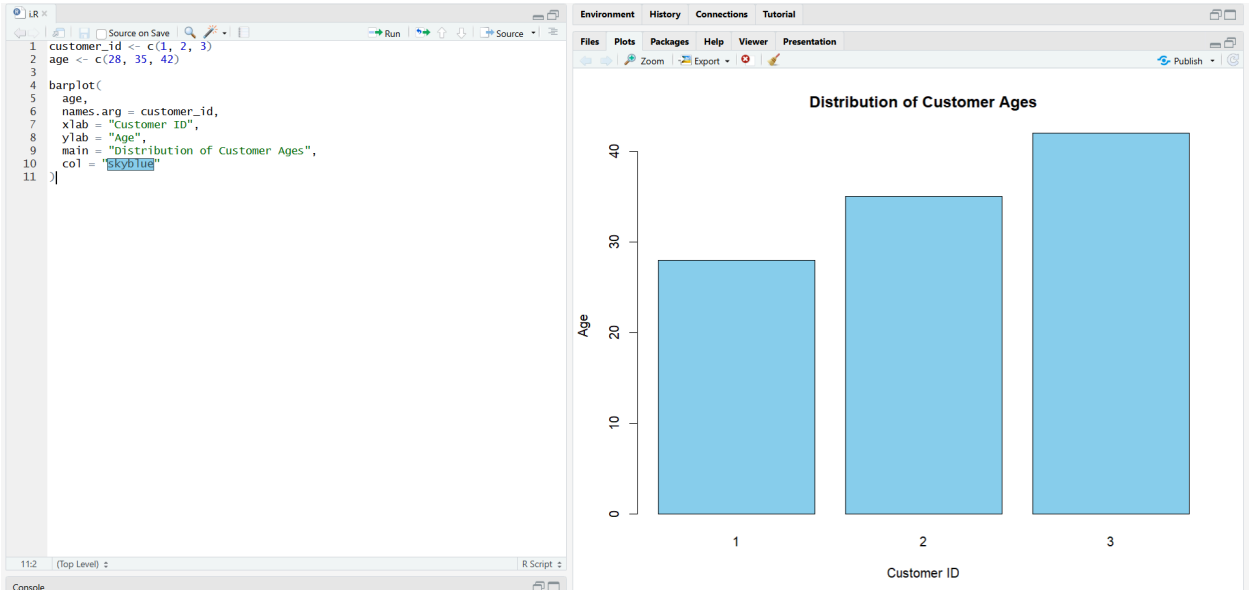
Customer ID	Age	Gender	Income ($)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

Q1. Create a bar chart to visualize the distribution of customer ages.

R Program

```
customer_id <- c(1, 2, 3)
age <- c(28, 35, 42)
barplot(
  age,
  names.arg = customer_id,
  xlab = "Customer ID",
  ylab = "Age",
  main = "Distribution of Customer Ages",
  col = "skyblue"
)
```

Output:



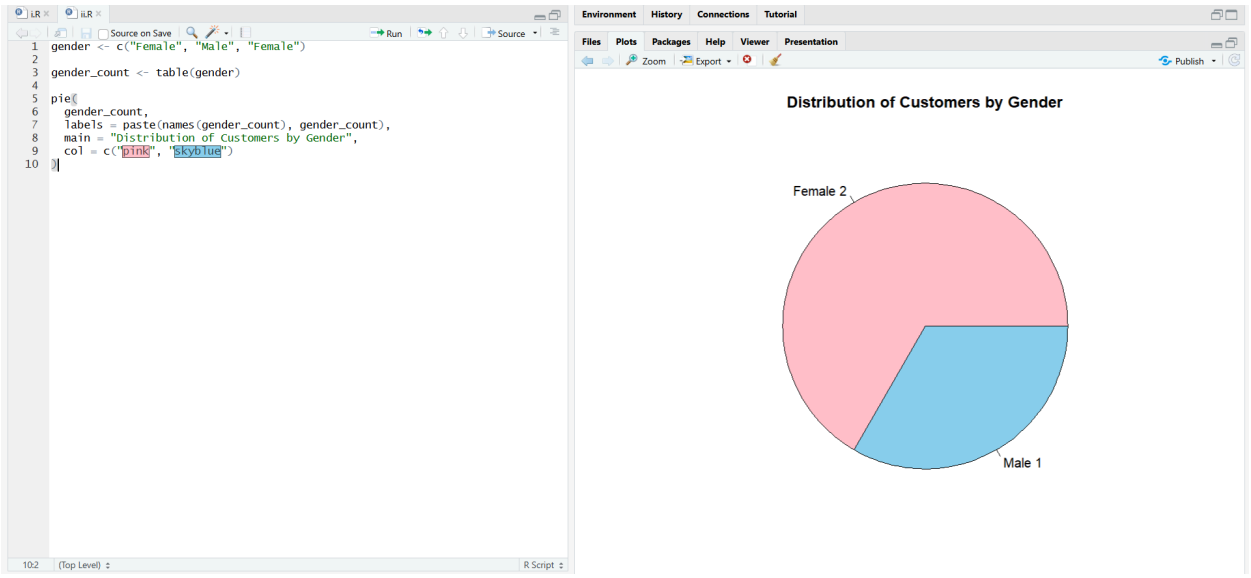
Observation: Customer 3 is the oldest at 42 years, while Customer 1 is the youngest at 28 years.

Q2. Generate a pie chart to represent the distribution of customers by gender.

R Program

```
gender <- c("Female", "Male", "Female")
gender_count <- table(gender)
pie(
  gender_count,
  labels = paste(names(gender_count), gender_count),
  main = "Distribution of Customers by Gender",
  col = c("pink", "skyblue")
)
```

Output:



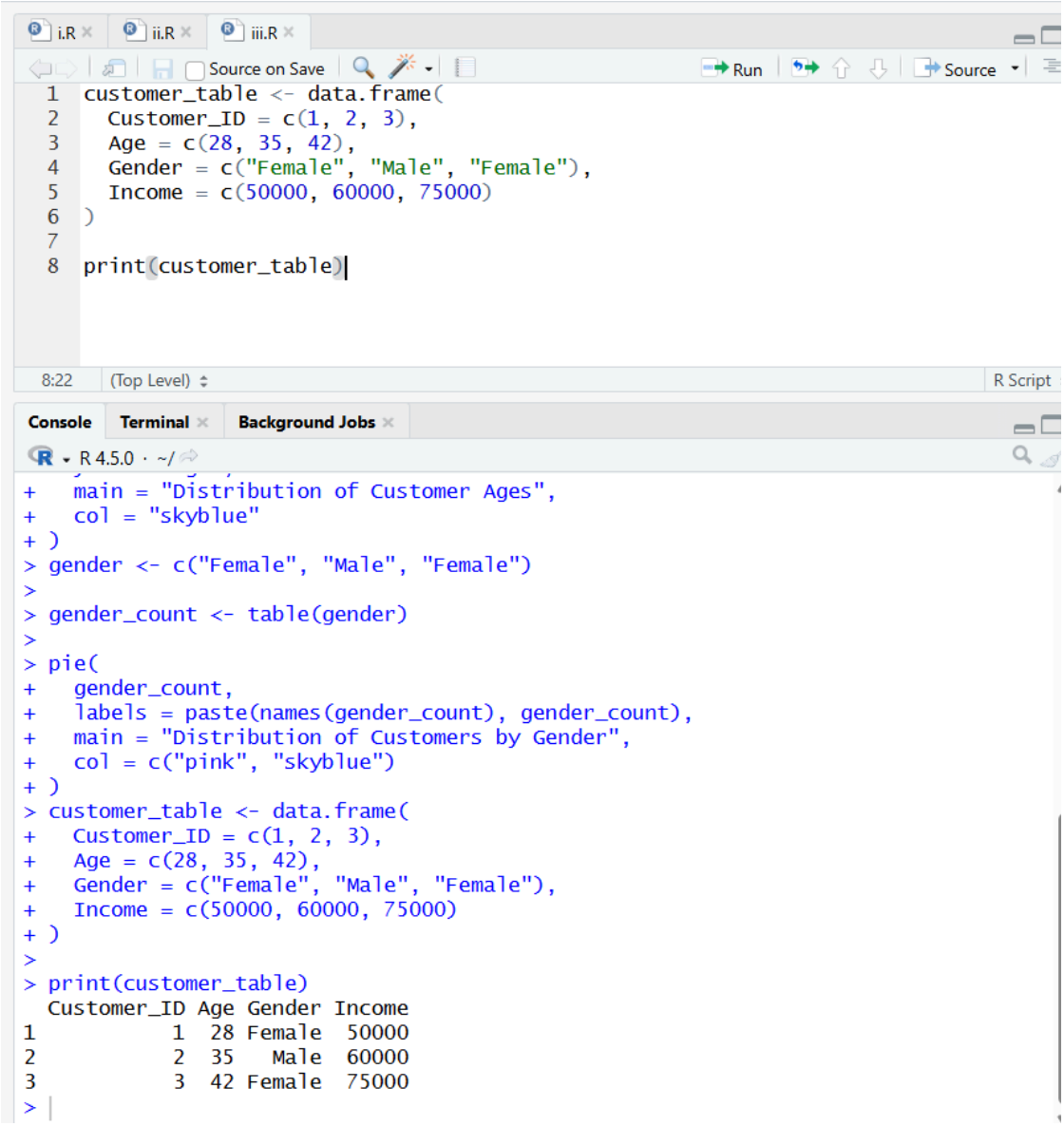
Observation: There are 2 female customers (66.7%) and 1 male customer (33.3%).

Q3. Build a table to show the demographic information for each customer.

R Program

```
customer_table <- data.frame(
  Customer_ID = c(1, 2, 3),
  Age = c(28, 35, 42),
  Gender = c("Female", "Male", "Female"),
  Income = c(50000, 60000, 75000)
)
print(customer_table)
```

Output



Observation: The table displays the age, gender, and income of each customer.

Q4. Develop a Tableau dashboard combining the bar chart, pie chart, and table.

Step 1 — Prepare the CSV dataset

Save the following as Customer_Demographics.csv.

Customer ID,Age,Gender,Income
1,28,Female,50000
2,35,Male,60000
3,42,Female,75000

Step 2 — Import the dataset

1. Open Tableau.
2. Select Connect → Text File.
3. Select Customer_Demographics.csv.
4. Open a new worksheet.

Step 3 — Create the bar chart

1. Drag Customer ID → Columns.
2. Drag Age → Rows.
3. Select Bar from the Marks dropdown.
4. Drag Age → Label to display ages.
5. Set the title to Distribution of Customer Ages.

Step 4 — Create the pie chart

1. Create a new worksheet.
2. Change the Marks dropdown to Pie.
3. Drag Gender → Color.
4. Drag Customer ID → Angle.
5. Change Customer ID to Measure → Count.
6. Drag Gender → Label.
7. Set the title to Distribution of Customers by Gender.

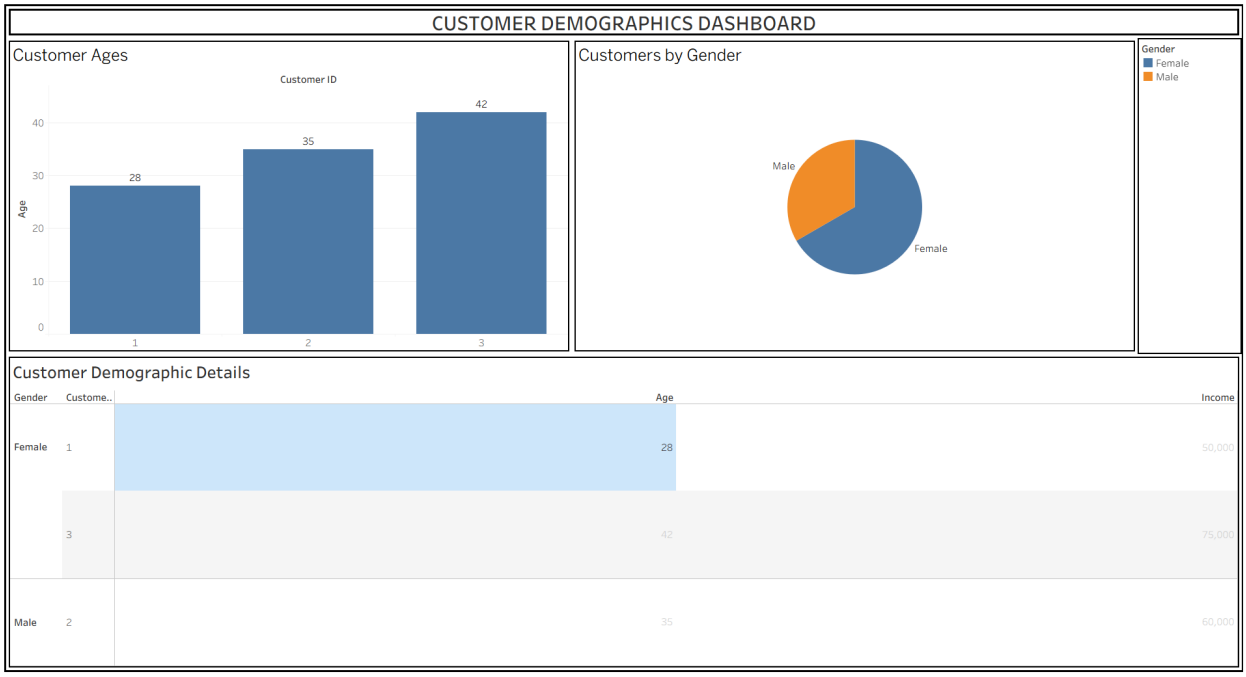
Step 5 — Create the demographic table

1. Create another worksheet.
2. Drag Customer ID → Rows.
3. Drag Age, Gender, and Income into the view.
4. Choose Show Me → Text Table if needed.
5. Set the title to Customer Demographic Details.

Step 6 — Create the dashboard

1. Click New Dashboard at the bottom.
2. Drag the Distribution of Customer Ages worksheet onto the dashboard.
3. Drag the Distribution of Customers by Gender worksheet beside it.
4. Drag the Customer Demographic Details worksheet below the charts.
5. Add filters such as Gender or Age, if required.
6. Arrange and format the dashboard neatly.

Expected dashboard



Procedure

1. Collected the customer demographic data.
2. Created a bar chart, pie chart, and demographic table using R.
3. Imported the dataset into Tableau.
4. Created the required visualizations and combined them into a dashboard.
5. Added filters to support interactive exploration.

Result

The customer demographics were successfully analyzed using bar and pie charts and a demographic table. A Tableau dashboard combining the three visualizations was designed for interactive exploration of customer data.